

Introduction

Bureau de change are currency exchange offices that provide foreign exchange services to consumers and businesses, typically involving the exchange of banknotes and travellers' cheques (IMF, 2016).

The foreign exchange market is marked by high liquidity and rapid transactional processing which makes it heavily regulated to monitor financial risks and combat financial crimes (money laundering, financial terrorism) (Reserve Bank of Zimbabwe, 2024). Consequently, Bureau de changes are required to integrate sound Know Your Customer (KYC) and due diligence procedures. This involves capturing and maintaining customer records. Many SMEs in developing countries are using spreadsheet-based systems to manually capture and process transactions. However, as the customer base expands, the risk of human error and security threats may rise since spreadsheet-based systems are more prone to data inaccuracies, and vulnerabilities which compromise operational resilience, potentially leading to business failure.

As someone who experienced the shortcomings of spreadsheet-based models firsthand, I saw the need for a more robust solution, which is why I chose to develop a BDC management system. This is an application designed to replace spreadsheet-based models with an efficient, secure and BI database-driven solution. The system supports real-time transaction processing and monitoring, automates key exchange rates calculations, tracks currency treasury balances, flags suspicious transactions and stores these records. Regarding the system cost model, the base database system (a databases dictionary with standard CRUD application) will be open source whilst a consolidated system with a Business Intelligence layer and a transaction monitoring system, will require a one-time perpetual license fee

Business Analysis

Identifying the primary users, their roles and how they interact with the system is very crucial to effectively design a database system that meet all end user requirements (Chen, 1976). The primary users of the system are:

Tellers

These are shopfloor employees who process customer requests at the booth. They use the system to log new transactions which involves customer onboarding, choosing currencies to be traded, inputting the forex amount received which enables the system to calculate the forex payout to the customer. They can view transaction history and can generate daily, weekly or monthly reports for managers. Tellers cannot alter exchange rates in the system.

Manager

The manager is responsible for the Bureau de change's profitability, treasury duties and overseeing daily operational activities. This involves setting the buy/sell exchange rates for all traded currencies, transaction monitoring and generating reports for key stakeholders. They resolve all system alerts raised which may involve lifting customer account holds triggered by transaction monitoring systems, and replenishing teller funds when treasury balances are low. They use the system's reporting dashboard to track key performance indicators: teller productivity, disbursement volumes, exchange rates and profitability ratios, among others.

Auditors/ Compliance officers

Their duty is to ensure integrity of all financial transactions by making sure the business unit adheres to micro finance regulations and internal policies. They verify client identification documentation and monitor transaction patterns for compliance with Anti Money Laundering regulations for Authorised Dealers with Limited Authorities (Financial Intelligence Unit, 2023). They do not process any transactions which is why they use the system in read-only mode capacity for data security reasons.

The system provides an immutable record for regulatory audits and a data warehouse for generating key compliance metrics for the business

Business Rules

The company policies, processes and standards of procedures translate to business rules, and these will be embedded within the databases design logic

1. Rate Application and Calculation

- for a customer selling foreign currency, a Teller initiates a transaction. The forex payout must be automatically calculated by the system, making use of the prevalent official Buy rate.
- Tellers cannot override the forex equivalent calculated by the system

2. Real-time liquidity management

- Prior to transaction completion, the system checks the CASH_INVENTORY to check if sufficient float is available for the currency to be disbursed.
- After each successful trade, the float balances for a pair of currencies involved are updated automatically.
- The transaction will be blocked if the teller is illiquid to settle the trade.

3. Immutable Audit Trail

- Transaction creation, exchange rate updates, user activity, among other actions in the system; must be logged in an AUDIT_LOG entity and the record must show an EmployeeID and a timestamp.

4. Single Active rate per currency

- This constraint will ensure that at any given time, only one EXCHANGE_RATE record per CurrencyCode is Active.

5. KYC enforcement

- capturing comprehensive KYC details is mandatory for creating customer records.

- For transactions exceeding the legally defined Domestic Utilisation of funds (equivalent of US\$500) (RBZ, 2024), the system holds the transaction if this information is incomplete or unverified.

6. Proactive Compliance and Sanctions Screening

- The system automatically screens all customers against an internal watchlist data dictionary. The watchlist data constituency include high-risk geographical regions (Iran, Pakistan), Politically exposed Persons, sanctioned individuals and companies (Office of Foreign Assets Control, 2023).
- If the system finds a potential match, a COMPLIANCE_FLAG is created, placing the transaction on hold and alerting the manager.
- The manager must review the flagged transaction and the system's match logical reason to make a data-driven decision; either to approve and complete the transaction or abort and report the transaction to a relevant regulating body.

7. **Markup calculation:** $\text{AppliedRate} = \text{OfficialRate} * (1 + \text{ManagerMarkup})$

8. Each **TRANSACTION** must be linked to a single **CUSTOMER** and an **EMPLOYEE** who processes it

9. Each **TRANSACTION** must capture the actual **EXCHANGE RATE** applied

10. **EMPLOYEES** must have an active status to process **TRANSACTIONS**, and inactive users cannot carryout teller operations

11. Only **EMPLOYEES** with “Manager” role can update daily **EXCHANGE RATES**, and any other **EMPLOYEE** cannot alter this rate

Table Design and Analysis

Based in the business requirements, the BDC system has 10 core data entities shown in Figure 1 namely:

1. Currency – stores standard codes and names for traded currencies

2. ExchangeRates – tracks current and historical exchange rates for various currency pairs
3. Customer – stores and maintains clients personally identified information and their KYC compliance record
4. Employee – stores system user information with role-based permissions
5. Role – defines user access levels and permissions which determines actions they can perform.
6. Transactions – Records all exchange transactions which includes parties involved and exchange rates
7. CashInventory – manages real-time currency float through tracking available balances and cash flows
8. KYCAuditLog – logs KYC verification history and results
9. ComplianceFlag – tracks suspicious transaction which includes flags raised and resolution status
10. AuditLog – captures system-wide activity which include user actions and data modifications

Entity Relationships

The ERD establishes crucial relationships (Chen, 1976):

Customer → Transaction (1: M) a customer can have multiple transactions, and each transaction is created by only one customer

Employee → Transactions (1: M) an employee can process multiple transactions, and each transaction is processed by only one employee

ExchangeRates → Transaction (1: M) A rate can be applied to multiple transaction, and each transaction can only have one exchange rate applied

Role → Employee (1: M) each role each role has multiple employees, and each employee has only one role

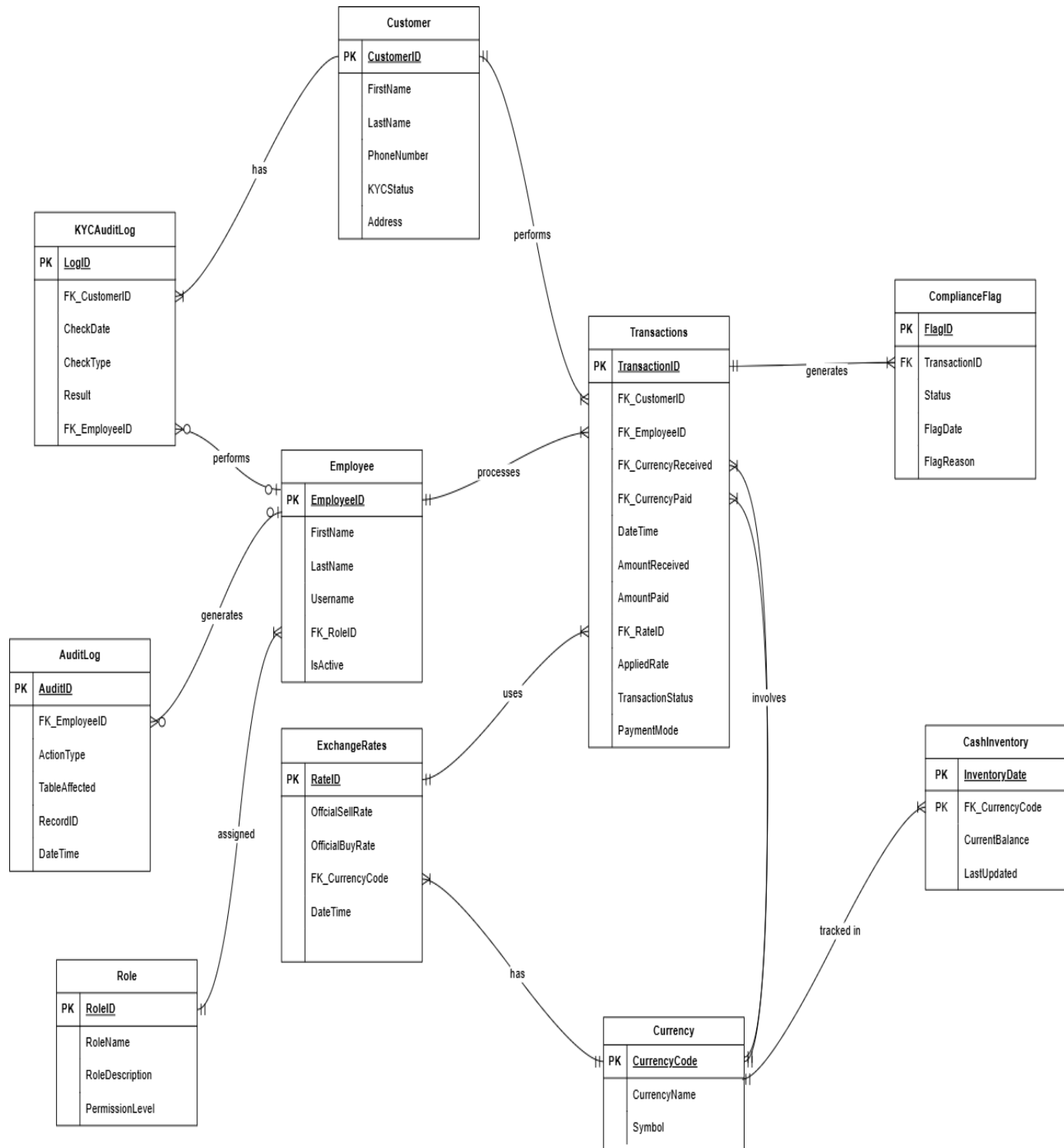


Figure 1: ERD Diagram for the BDC Management system

Transaction → ComplianceFlag (1:1) each flagged transaction has one flag record and each flag record is associated with a single transaction

Currency → ExchangeRates (1: M) each currency has multiple rate records, and each rate record is quoted in a single currency

Database Implementation

Scenario 1: Customer Onboarding and Initial Exchange Transaction

Step1: *Customer Registration and KYC Check*

A new customer, Beefy Jogger visits a branch and meets Teller 5. The customer signs up for service, providing details and completing a KYC check

Step 2: *KYC Verification*

Teller 5 completes the KYC verification process and the customer's status is updated to "Verified"

Step 3: *Exchange Transaction*

The customer initiates a USD to ZAR exchange transaction worth \$1000 and teller 5 processes the transaction. A KYC status check is initiated before processing this large transaction.

Step 4: *Cash inventory and log transaction update*

After the transaction, cash inventory balances and audit logs are automatically updated

	SQL 1*	SQL 2*
1	--Customer Onboarding and Initial Exchange TRANSACTION	
2	--1. Create new customer with KYC check	
3	INSERT INTO Customer(	
4	CustomerID, FirstName, LastName, PhoneNumber, Address, KYCStatus)	
5	VALUES(73, "Beefy", "Jogger", "+16147477376", "123 Albion St, Boston", "Pending");	
6		
7	-- Log KYC verification	
8	INSERT INTO KYCAuditLog(	
9	LogID, FK_CustomerID, CheckDate, CheckType, Result, FK_EmployeeID)	
10	VALUES(64, 73, CURRENT_TIMESTAMP, "DocumentVerification", "Pass", 5);	
11		
12	-- Update customer KYC Status	
13	UPDATE Customer	
14	SET KYCStatus = "Verified"	
15	WHERE CustomerID = 73;	
16		
17	-- 2. Verify customer KYC status before processing large transaction	
18	SELECT t.TransactionID, c.CustomerID, c.FirstName, c.LastName, c.KYCStatus, t.AmountReceived,	
19	CASE	
20	WHEN t.AmountReceived > 500 AND c.KYCStatus != "Verified"	
21	THEN "Hold"	
22	ELSE "Proceed"	
23	END as TransactionAction	
24	FROM Transactions t	
25	JOIN Customer c ON t.FK_CustomerID=c.CustomerID	
26	WHERE t.TransactionID= 63;	
27		
28	-- 3. Process USD to ZAR exchange transaction	
29	INSERT INTO Transactions(TransactionID, FK_CustomerID, FK_EmployeeID, FK_CurrencyReceived,	
30	FK_CurrencyPaid, DateTime, AmountReceived, AmountPaid, FK_RateID, AppliedRate,	
31	TransactionStatus, PaymentMode)	
32	VALUES(63, 73, 5, "USD", "ZAR", CURRENT_TIMESTAMP, 1000, 16000.00, 20, 16.00, "Completed", "CaSH");	
33		
34	-- 4. Update cash inventory after disbursement	
35	-- Log transaction in audit trail	
36		
37		

	TransactionID	CustomerID	FirstName	LastName	KYCStatus	AmountReceived	TransactionAction
1	63	73	Beefy	Jogger	Verified	1000.0	Proceed

Figure 2: Customer onboarding and initial transaction process

Analytics, Reports and Metrics

The database contains information that can be leveraged to make data driven decisions, optimize operations and ensure compliance with regulatory requirements. The following analytics and reports can be generated for insights gleaned from the database:

Transaction Status Report

This report provides an overview of the current state of transactions which includes the number of completes and pending transactions, annotated with corresponding percentages

SQL 1*			
1	-- Completed vs Pending Transactions		
2	SELECT		
3	TransactionStatus,		
4	COUNT(*) as Counts,		
5	ROUND(COUNT(*) * 100 / SUM(COUNT(*) OVER(), 2) as Percentage		
6	FROM Transactions		
7	GROUP BY TransactionStatus;		

	TransactionStatus	Counts	Percentage
1	Cancelled	5	9.0
2	Completed	23	45.0
3	Flagged	8	15.0
4	Pending	15	29.0

Figure 3: Transaction status table

Employee Performance Report

This report evaluates employee productivity, measured by the number of transactions completed by each employee, arranged in descending order

SQL 1*			
1	--Employee Transaction Count		
2	SELECT e.FirstName, e.LastName, COUNT(t.TransactionID) as TransactionCount		
3	FROM Employee e		
4	LEFT JOIN Transactions t ON e.EmployeeID=t.FK_EmployeeID		
5	GROUP BY e.EmployeeID		
6	ORDER BY TransactionCount DESC;		

	FirstName	LastName	TransactionCount
1	Paul	Sams	12
2	Brian	Taylor	11
3	Anna	Mendoza	11
4	John	Compliance	7
5	Clara	Chivonyl	7
6	Mary	Chewbacca	3

Figure 4: Employee performance table

Flagged Transaction Report

This report identifies transactions that have been flagged for review, which allows for further investigation and potential resolution

SQL 1*					
1	--Flagged Transactions for Review				
2	SELECT t.TransactionID, cf.FlagDate, cf.FlagReason, cf.Status, t.AmountPaid				
3	FROM ComplianceFlag cf				
4	JOIN Transactions t ON cf.FK_TransactionID=t.TransactionID				
5	WHERE cf.Status = "Active";				

	TransactionID	FlagDate	FlagReason	Status	AmountPaid
1	34	2025-11-10 17:35:00	Unusual currency pair and pattern	Active	2147.03
2	20	2025-11-10 14:05:00	Customer on internal watchlist	Active	2665.18
3	1	2025-11-10 09:20:00	Customer on internal watchlist	Active	839.51
4	19	2025-11-10 13:50:00	Customer on internal watchlist	Active	742.26
5	38	2025-11-10 18:35:00	Multiple transactions just below ...	Active	4454.76
6	25	2025-11-10 15:20:00	Multiple transactions just below ...	Active	2266.38
7	2	2025-11-10 09:35:00	Multiple transactions just below ...	Active	1481.46
8	26	2025-11-10 15:35:00	High value transaction above daily ...	Active	4495.43

Figure 5: Flagged transactions overview

Exchange Rates report

This report displays the current exchange rates for all currencies traded, as mandated by regulatory requirements for Bureau de Change operators. These will be made available to customers for viewing through our digital dashboards

Security Concerns: Data Protection and Compliance Requirements

Financial data sensitivity – Sensitive information from transaction records will be protected under the financial privacy regulations (RBZ, 2024). This includes customer financial profiles, business turnover data and exchange amounts

KYC Data Protection – Personally identifiable information (PII) which comprises of customer identification documents, addresses and KYC profiles, will be encrypted at rest. Strict access control will also be integrated in the system

Transaction monitoring – A secure API integration with encrypted data transmission will be used for real-time screening against sanction lists (UN, EU) and Politically Exposed Persons (PEP) databases (Financial Action Task Force, 2022)

Data Retention Compliance: An automated archival process with encrypted long-term storage will be implemented to meet the RBZ's requirement for BDCs to maintain transaction records for 5 years.

Multi-factor Authentication: For transactions exceeding the Domestic Utilisation threshold, the system should require Multi-Factor Authentication from both the manager and teller (National Institute of Standards and Technology, 2020).

Architecture

The technical requirements for a Bureau de Change Management database system will involve a three-tier Client-Server Model deployed using a Hybrid cloud Approach and significantly large storage capacity to store customer and transaction records for at least 5 years (Microsoft Azure, 2023)

System Architecture:

Three-Tier Client-Server Model:

1. *Client*

Tellers will have access to desktops to process transactions. Mobile applications will be made available to managers for them to set trading rates, resolve raised compliance flags and have summary statistics of the branch performance. Web portals for database administrators will be integrated for ease use and management of the database system

2. *Server*

The application server runs the business logic and enables smooth handling of user authentication, authorization and transaction processing. Sensitive customer information

(eg. addresses) will be secure enough since this design enables improved security and maintainability of customer record

3. *Database*

A relational database represented in the ERD below will be implemented. To ensure business continuity, backup and replication systems which duplicate real-time data will be integrated in the system. User system activity will be monitored by the audit logging system to provide temper-proof records for regulatory compliance.

Deployment Strategy

Hybrid Cloud Approach:

The infrastructure consists of a primary cloud-based Amazon RDS for PostgreSQL and Amazon RDS for SQLite for offline capability during internet outages. Cloud backups denormalized to capture geographic data redundancy will be in place to ensure business continuity in times of data corruption disasters. Daily data synchronization process between the central database and branches will be initiated to enhance data availability across all branches (Amazon Web Services, 2023).

Storage Requirements

Assuming the BDC operations typically process 200-500 daily transactions, storage needs are moderate but must account for a 5-year retention period. Transaction data might require 10 GB of storage, if 50,000 transactions at 200 KB each, will be processed during the first year. I will set aside 2 GB for customer records which will be enough to cater for 5000 customers KYC documents. Audit logs, compliance data and exchange rate history will be spared 15 GB, 2GB and 1GB of storage, which brings the total initial storage to 30GB. The system is expected to experience a 25% annual growth rate, reaching peak capacity of 250 GB by year 5, with an additional 80GB of backup storage allocated for weekly and monthly back-ups

Project Wrap-up and future Considerations

This database design project provides a solution to error prone spreadsheet-based systems which balances operational efficiency and regulatory compliance in financial systems. Through this exercise, I learnt the importance of a thorough database design prior to implementation, the need to understand end-user requirements (business rules) and SQL skills to translate those requirements into functional database system (Connolly & Begg, 2015). The design blueprint provides an overview of the problem statement which aid application programmers and other stakeholders to have a unified approach when implementing a database system. Future consideration for this project include implementing additional data validation constraints, adopting and Api-first architecture, integrating predictive analytics for cash flow forecasts and leveraging machine learning for risk scoring.

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